



Facial action unit detection methodology with application in Brazilian sign language recognition

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Abstract

Sign Language is the linguistic system adopted by the Deaf to communicate. The lack of fully-fledged Automatic Sign Language (ASLR) technologies contributes to the numerous difficulties that deaf individuals face in the absence of an interpreter, such as in private health appointments or in emergency situations. A challenging problem in the development of reliable ASLR systems is that sign languages do not rely only on manual gestures but also on facial expressions and other non-manual markers. This paper proposes to adopt Facial Action Coding System to encode sign language facial expressions. However, the state-of-the-art of Action Unit (AU) recognition models is mostly targeted to classify two dozen of AUs, typically related to the expression of emotions. We adopted Brazilian Sign Language (Libras) as our case study and we identified more than one hundred of AUs (with a great intersection with other sign languages). We then implemented and evaluated a novel AU recognition model architecture that combines SqueezeNet and geometric-based features. Our model obtained 88% of accuracy for 119 classes. Combined with the state-of-the-art of gesture recognition, our model is ready to improve sign disambiguation and to advance ASLR.

Keywords Facial expression · Non-manual markers · AU detection · Libras · Brazilian sign language

1 Introduction

Automatic Sign Language Recognition (ASLR) systems are designed to process, recognize and interpret sign language sentences performed by an individual in a video, and to transcribe them to text or audio in a target language. Associated with sign language synthesis technology, ASLR has the potential to improve the interaction between deaf individuals and oral language speakers, facilitating situations in which the presence of an interpreter is not desirable or not possible, such as in private health appointments or in emergency situations. Such assistive tools are also considered key technologies to improve the access of deaf individuals to education and to health information [13].

ASLR systems' development faces three major challenges: First, a few sign language annotated corpora are available. As a visuospatial linguistic system, Sign Languages (SL) vary worldwide, and they can be typically characterized as a minority understudied language. Second, transcribing sign language into a written system is not trivial, making it challenging to create new corpora. Also, the existing corpora do not present a model transcription pattern, making it difficult to reproduce. Lastly, the recognition of sign language patterns represents a challenging computer vision problem. SL recognition does not involve only the accurate detection, tracking, pattern analysis of the handshapes, the arms' position, and the hands' trajectory. In many cases, to disambiguate signs or define a sentence's meaning, it is necessary to look to the face.

In sign languages, facial expressions are part of the grammatical structure of the language. Some facial expressions represent morphemes that fulfill syntactic and pragmatic functions. When a facial expression is not employed, the message can change or be misunderstood, causing communication difficulties [20, 58].

However, the recognition of SL facial expressions is more complicated than the typical affective computing problem

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Fig. 1 Examples of facial expressions in sign language that are not associated to the expression of emotions. In the image, the Libras' signs for "Motel" and "Skinny" are shown, respectively. Image extracted from [59]

since the variety of sign language facial expressions is significantly greater than the facial expressions of emotions, as illustrated in Fig. 1. In the literature, few works approach the theme, and state-of-the-art facial expression recognition (FER) models do not generalize well when tested on new assignments, nor can they easily be adapted to new facial expressions without a substantial number of ground truth labels [36] (citation removed for blind review).

We follow the same approach of existing FER models that adopt the Facial Action Coding System (FACS) to encode facial expressions in this work. FACS defines Action Units (AUs) that are codes for non-overlapping anatomically-based facial muscle movements [19]. Since AUs are based on coding the facial muscle movements, it is neutral regarding whether it is coding linguistic or affective facial actions. Additionally, FACS specifies Action Descriptors (AD) and visibility codes. ADs describe actions that have no muscular basis, like swallowing, cheeks blowing, and others. Visibility codes are labels for occlusion or undefined motion [18]. When combined, FACS can describe any facial expression, and further interpretations of them can be inferred from it.

In particular, we adopt the Brazilian Sign Language (or Libras) as our language of study. In previous works, we described our efforts to study and to encode Libras' facial expressions using FACS, and to build the first annotated corpora using FACS for Libras (citation removed for blind review). This trajectory helped us to identify the great variety of facial expressions in Libras, and to understand their grammatical role.

In this work, we propose a novel AU classifier model, with application in ASLR (and other potential applications). The proposed model fuses region of interest and facial geometrical information with existing deep network architectures to be more accurate in the recognition of action units. While state-of-the-art AU classification models only handled eight to twenty AU labels [10, 39, 53, 79], our experiment involved 119 AUs that we identified for Libras, without

major loss of performance. Being more general, our results are comparable to the literature results. Furthermore, the novelty arises from the use of 14 AUs, never previously used in an AU recognition system, much less in SL's analysis.

Thus, we propose two novelties here, the combination of the proposed preprocessing with the SqueezeNet model and its application in a wide variety of facial action units never used in FER and SLR studies.

The paper is further organized as follows. Section 2 reviews prior works on AU classification. Section 3 provides a brief revision of Libras' facial expressions and their labels. Section 4 brings a technical description of the method proposed for AU classification. Section 5 has a dataset description and implementation details. Section 6 presents the performance achieved by the system, an ablation study, and details of the conducted cross-dataset experiments. Finally, the last section concludes the paper.

2 Related works in AU detection

In conventional AU recognition methods, the face and the facial landmarks are detected, then the features are extracted from the face, and, finally, multiclass machine learning models, such as Convolutional Neural Networks (CNN), classifies the AUs [3, 39, 52, 55, 71].

Seen as a crucial step, typical approaches for the feature extraction stage can be divided into geometric-based and appearance-based approaches. The former uses the landmarks or geometric properties of a face, i.e., location of facial points, such as eyes, nose, mouth and the distance between those facial points and the shape of facial components [17, 28, 67]. While the appearance-based approaches use pixels intensity values as characteristics, i.e., the information goes directly from face images into the recognition algorithms [21]. With the development of Convolutional Neural Networks, the techniques to extract local features appeared [47]. The face is first registered in the extraction of local features step according to some landmark detection algorithm and further cropped independently in certain regions. Such models have a Region of Interest (ROI) selected, and then, CNN-based features can be extracted independently. Based on the feature region scheme, some works choose to build on top of region layers. Li et al. [38] introduced the CNN-based ROI detection, which was then incorporated into the local modeling of AUs. They proposed a fully-connected LSTM network to exploit the temporal correlation fed with independent ROI networks to learn separate filters for different facial regions. Proposed by Li et al. [39], the EAC-net tries to enhance and crop regions of interest around facial landmarks accordingly with AU active regions. The heuristic nature by which attention masks are created to enhance and crop the ROI is one limitation of

this approach. Shao et al. [54] introduced a framework that jointly performs facial action unit detection and face alignment, where a significant improvement in AU detection was observed.

Recently, a vast number of deep learning architectures have been proposed for specific tasks, and CNNs are the most popular tool for AU detection [9, 80]. In Gudi et al. [22], the authors used a seven-layer CNN for estimating AU occurrence and intensity, and their results suffered from potential overfitting. Extensive CNN experiments were done in [50], where 23 CNN models were used to identify 20 AUs and 3 AU combinations. Their architecture achieved an accuracy rate above 90% for both AUs and emotions. Yet, for AU detection, they only trained and tested on a small dataset, the Extended Cohn-Kanade Dataset (CK+) [29]. Jia et al. [27] proposed a multi-layer fusion system based on AUs with a stacking framework, where the association rules employed were used to explore and find the relationships between AUs and facial expressions. Velusamy et al. [68] infer facial expressions from AUs sequences by submitting a Longest Common Sub-sequence (LCS) distance to match the AUs strings. Recent works that attempted a hybrid strategy, still try to learn a global representation from the input image where AU relations are combined through either generative or discriminative approaches [24, 51]. By taking advantage of spatial CNNs and temporal LSTMs, Chu et al. [10] presents a fusion hybrid network with performance on the BP4D dataset [77] over 12 AUs, and in a tenfold protocol, an average F1 score of 82.5.

All of these works require, however, preprocessing step that consists of registering the input image according to some detected facial landmarks and further alignment. In this paper, we observe that both feature schemes (geometric landmark aided and region of interest) can be performed together in a rather simple way. While the methodology proposed in this paper is entirely new, some works have attempted to implement facial landmarks to improve the performance of AU modeling in a unified framework [66, 81].

Furthermore, other works bring the AU semantic relationship through the network using statistical calculations based on anatomical information, since among different AUs are strong associations under different facial expressions. Liu et al. [41] use the spectral perspective of Graph Convolutional Network (GCN) in AU detection. They demonstrate a strong learning ability between the defined four-module AU graph structure and the optimization network. Another approach contemplates the difference between the apex of the expressive face and its corresponding neutral face a valid prior knowledge [7, 31, 75]. Most of these works mentioned so far are emotion-specific FER systems based on AUs detection, which may consist of facial action unit detection and further inference of emotional facial expression labels. Few researchers are focused on the indirect use of AUs in

advanced application systems due to the inherent challenge of AU detection. Particularly in the SL field, Dachkovsky and Sandler [12], and De Vos et al. [14] associated non-manual markers with FACS, and Silva et al. [58, 59] proposed a scheme for recognition of facial expression in SL by using AU detection. However, recognition of non-manual markers is generally treated as some type of written form of the signs or the authors create their classes of facial actions [74]. A comprehensive survey of AU analysis can be found in [42]. The classification of non-manual cues in computer vision issues is related to extracting facial features, eye gaze, and head pose cues. Also, since there is no complete and standardized written form for SLs articulators, there exists an inevitable range of different approaches to describe them, making a problematic trade-off between recognition accuracy and generality onto various SLs applications.

Moreover, AU recognition can also be applied in pain studies, like in [73] where they analyze the relationship between sequence-level multidimensional pain measurements and an AU derived pain-related measure, by applying a multitask learning model to predict the visual analog scale from human-labeled AU. In intelligent vehicle systems, facial motion and appearance changes occurring during drowsiness can also be obtained from detection and recognition of related AUs [70]. Thereby, the sets of AU used in such studies may not be the same as the set of emotion facial expressions used on most works. In contrast, we assume an extensive set of AU with an approach that fused texture and geometric information at the data level by mapping the geometric descriptors onto graphic heatmap curves, then rendering multiple cropped face images, which are fed as input to the backbone network model. The output will predict AU presence for a Libras application.

3 Libras' facial expressions: labels and features

In Brazil, Libras was recognized by the legal system as a linguistic system in 2002 [5]. As a sign language, Libras is grammatically structured and organized into a word and sentence-forming system [65]. By definition, signs are structured with simultaneous parameters that form two categories, the Non-Manual Markers (NMM) composed of facial and body expressions, and the Manual Markers (MM) containing the other hand-expressed signals [1, 48, 76]. Facial expressions as an NMMs operate as indicators that complement the discourse with semantic, lexical, and prosodic information, making their study relevant [6]. In addition to idiolect-related expressions, sign language facial expressions provide affective information, and they are part of non-verbal communication.

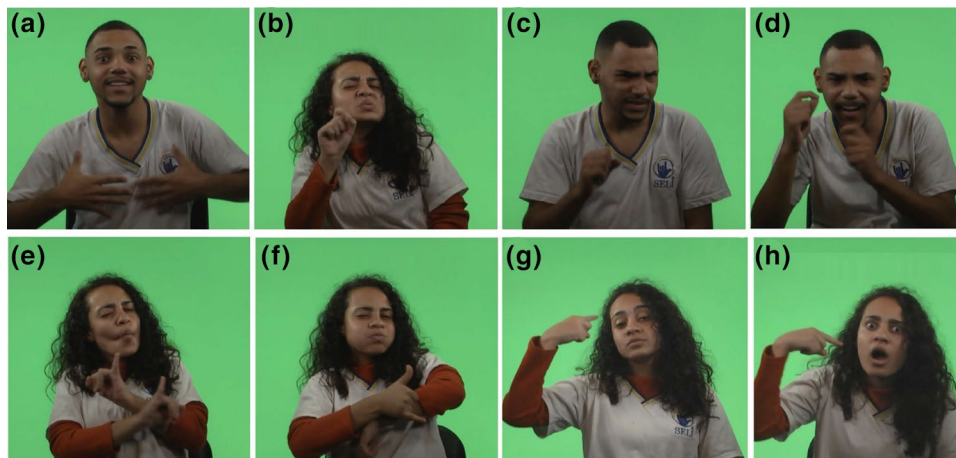


Fig. 2 In the performance of the signs in Libras, we can analyze the variation of the facial expressions by the images (a)–(e). Image (a) is an example of an affective facial expression by the interpreter signing “Joy”, which is accompanied by a smile and raised eyebrows. The grammatical facial expression of sentences (GES) is subdivided accordingly with their syntactic function as WH-Question, YN-Question, Doubt, Negative, Assertive, Topic, Relative, and Conditional Clause. Images (b) and (c) portrayed examples of GFE of sentences for WH-Question and Doubt, where the portrayed signs are “Which?” and “Can?”, respectively. Conveyed by image (d) is the sign “Dental floss,” which has a facial expression by definition, and it is called a

GFE of norm. Moreover, in images (e) and (f), the interpreter performed the sign “very slim” and “very fat,” respectively. The intensity of the sign is displayed by the change in the facial expression, which passed from neutral to frown and sucked cheeks (e), or to frown and inflated cheeks (f). Those are examples of GFE of intensity. Defined as GFE of homonymy are facial expressions that are performed with the same manual gesture. In the last images (g) and (h), the signs “lawyer” and “crazy”, where we can observe that their difference is only based on the eyes, eyebrows, and mouth action. All images are from SILFA dataset [58, 59]

Although there may be different ways of classifying the signs that make up the Libras lexical unit, the mapping of linguistic facial expressions can be done by separating the type of function and the meaning that the expression plays in the sentence. Indeed, when facial expressions transmit the grammatical information in a sentence, they are called grammatical facial expressions (GFE). In [58, 59], it was proposed that grammatical facial expressions vary accordingly with their function and can be divided into four different classes: GFE of sentence, intensity, homonymy, and norm. Moreover, there are facial expressions that carry emotion or an affective state, which is called affective facial expressions (AFE). The linguistic and affective marks of facial expressions differ in SL in several ways, and that marks the different use of the same facial muscles [58, 59]. In Fig. 2, we show examples of an AFE and each kind of GFE.

Included in the GFE class are the classifiers of quantity and size, and these carry the most significant number of non-manual articulators that are excluded in FER works. The particularities of linguistic structures (classifiers, depicting signs, shift roles, storytelling, among others) and the iconicity present in natural discourses of Libras for non-manual articulators, in addition to an in-depth literature review, can be found in detail on studies done by Silva [60].

Despite the given description of facial expressions in the Libras scheme is exceptionally detailed, they are not machine-readable. To fulfill such a requirement, Libras’

FEs are associated with the facial action unit coding in [58, 59]. A major advantage to adopt FACS to describe FEs in Libras is the possibility to unify and standardize a transcription method, which is heavily based on consolidated background theory, has available resources of annotation training, and is also applied in other fields, turning it possible cross studies. Based on the type of meaning and function that the FE plays in a sentence, our approach defines simple and compound facial expressions. Here, simple facial expressions (or simple NMMs) are those that use only one action unit in their performance. In comparison, compound facial expressions (or compound NMMs) integrate the combination of simple FE components in their performance. For each definition, an example is given in Fig. 3.

In Santos and Xavier [15], it was identified that 42% of the Libras’ signs analyzed in his corpus used only one articulator (with mouths and cheeks being the most used), but that 58% use more than one articulator. Additionally, in Silva et al. [58, 59], it is described 39 simple facial actions that can be combined into more than 80 compound facial expressions, that show the comprehensiveness of Libras’ FE. Based on these works, Table 1 presents associated codes to the lexical and grammatical facial expressions in Libras. Also, Table 1 exhibits the possibilities of action units that are typically associated with the expression of basic emotions and widely adopted in computational studies.

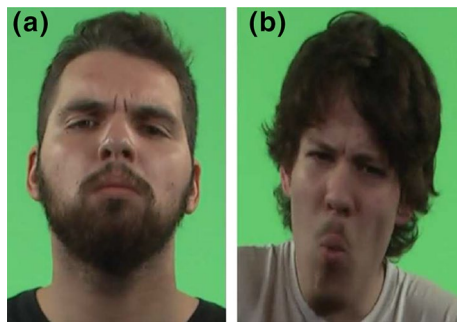


Fig. 3 Examples of simple and compound facial expressions. In Image (a), an example of a simple facial expression by the portraying of “frown” where only AU4 is necessary to represent the eyebrows lowered and pushed together. In Image (b), an example of a compound facial expression for the combined performance of “frown, projected lips and head tilt to the side” labeled as AU4 + AU26 + AU23 + AU55. Images are from [58]

With the features and labels described in Table 1, by comparison, Libras uses 26 FACS AUs, while emotion studies use 16 FACS AUs. More than that, only 4 FACS AUs of basic emotion (AU 7, 9, 14, 20) are not involved

in Libras’ facial expressions. That means 14 FACS AUs (AU 18, 19, 22, 24, 33, 34, 34R, 35, 44, 46, 51, 52, 53, 54) are not usually employed in studies of facial expression or even included in action unit analysis. In other words, by studying facial expression in sign language, we were able to transcend the work on action units to those that usually are neglected.

Note that FACS encodes any facial movement by a description of small unit codes or combination thereof. So, as applied in previous works (reference removed for blind revision), the usage of facial action units is done to describe the facial muscular movements more accurately than in terms of non-manual signs (that enclose both facial expressions and body movements). For example, the action unit from non-manual signs in Fig. 2(c), and Fig. 2(d) usually receives the same transcription in SLR analysis, i.e., ‘frown and open mouth’. At the same time, in FACS, they would bring the annotations for the precise muscular movement related to the mouth, i.e., AU4 + 16 + 25 and AU4 + 12 + 26. Such exact clarification introduces more information to the recognizer.

Table 1 Set of action units related with Libras’ execution in discourse

Description of facial expression	Involved Action Units
<i>Affective facial expressions</i>	
Surprise	AU 1, 2, 5, 15, 16, 20, 26
Fear	AU 1, 2, 4, 5, 15, 20, 26
Disgust	AU 2, 4, 9, 15, 17
Anger	AU 2, 4, 7, 9, 10, 20, 26
Happiness	AU 1, 6, 12, 14
Sadness	AU 1, 4, 15, 23
<i>Grammatical facial expressions</i>	
GES WH-Question	AU 4, 18, 23, 53
GES YN-Question	AU 1, 2, 4, 18, 23, 51, 52, 53
GES Doubt	AU 4, 24, 28, 44
GES Topic	AU 1, 2, 5, 18, 22, 23, 25, 26, 53, 54
GES Negative	AU 4, 15, 17, 51, 52, 54
GES Assertive	AU 53, 54
GES Focus	AU 1, 2, 5, 18, 22, 23, 25, 26, 53, 54
GES Condicional Clause	AU 1, 2, 53
GES Relative Clause	AU 1, 2
Grammatical expression of iIntensity	AU 4, 10, 12, 25, 26, 28, 33, 34, 35, 44, 53, 54
Grammatical expression of Homonymy	AU 10, 12, 18, 23, 25, 26, 34, 34R, 46
Grammatical expression of norm	AU 1, 2, 4, 12, 15, 16, 17, 18, 19, 28, 33, 34, 35, 51, 52, 53, 54
<i>Appearance labels</i>	
Neutral	AU 0
Face occlusion	AU 71, 72, 73
Unscorable	AU 74

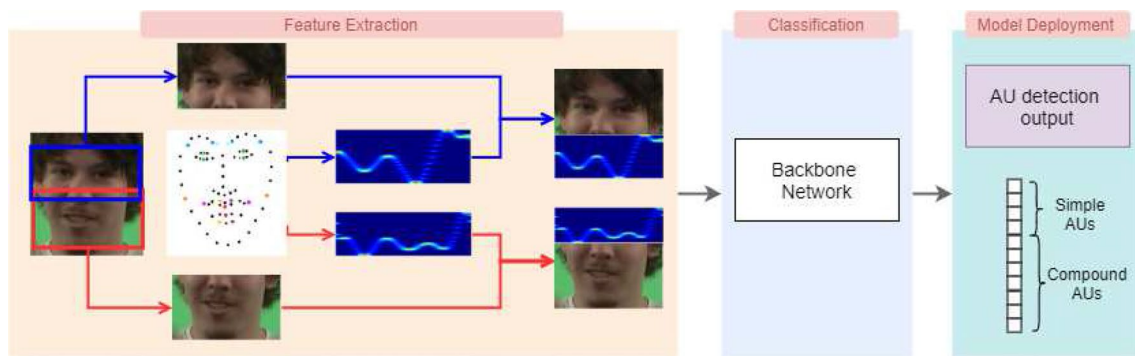


Fig. 4 Overview of the proposed pipeline for Libras' AU detection

4 Proposed approach

In this section, we present a novel scheme for AU classification using three steps, as shown in Fig. 4. We start with the detection of the region of interest and geometric features, followed by a deep neural network choice. The main novelty of this framework lies in the transformation of geometric features, allowing the network to gain a spatial aspect. By using a region of interest approach, we can gather elements at different positional levels, which provides the network capability to learn both the spatial relation and co-occurrence of AUs. Our approach applies multi-class learning through estimation of a combination of AUs, in the sense that AUs' spatial relation and co-occurrence is inherently embedded in the labeling method.

4.1 Region of interest

Detecting the position of a face on an image may be a challenging task due to rigid (scale, rotation, and translation) and non-rigid (such as facial expression variation) face transformations. A robust algorithm capable of learning not only the position of the face but also indicating the location of facial elements is ideal. One of the most popular robust face detectors with great accuracy is a regression-based method constructed by a classic HOG feature combined with a linear classifier, an image pyramid, and a sliding window detection scheme (Haar-Cascade features) [30, 69]. The result of this development can be found in Dlib [32] and OpenCV [26] libraries and was applied in our build.

After the detection of the face in the image, a fixed bounding box is drawn around and cropped from the picture. Then, the new image containing only the face is resized into 96 by 96 pixels. Although many datasets are composed of higher-resolution images, tests suggest that decreasing the resolution does not impact accuracy, and it increases the speed of the network [46].

For the feature extraction part, the face image is cropped into two ROIs, a lower portion of the face comprehending

the chin, mouth, and nose, as well as the upper portion of the face embracing the forehead, eyebrows, and eyes. Our region of interest was chosen based on Libras linguistic studies [58]. The face articulators are divided into two categories, the ones belonging to the upper face part and the others that belong to the lower face part. The resultants sectioned images have a 60×96 size for each cropped face part, with an overlapping as shown in Fig. 4.

4.2 Geometrical features

Based on the advantages of a hybrid face expression detection approach, we also extract the position of defined facial points. We propose to fuse the shape information with the landmarks euclidean distance in terms of multiple descriptors by interpolating a define curve. Using the Dlib implementation of this Haar-Cascade method we extracted 68 landmarks localized on the face placed alongside the ears, chin, nose, eyes, eyebrows and, mouth. This landmarks we denoted by $l_i = (x_i, y_i)$ where $i \in \{1, \dots, 68\}$.

Since AUs are measures defined as changes in facial configuration, by calculating the distance between key points in the eyes or mouth, is possible to infer if they are open or closed. Similar shape information can be obtained for eyebrows, cheeks and lips. So, we composed points vectors with face landmarks l_i , $i = 1, \dots, 68$ and the euclidean distance measures $d_2(l_j, l_k)$ with $(j, k) \in \{(3, 13), (17, 21), (21, 22), (22, 26), (38, 40), (43, 47), (48, 54), (50, 58), (52, 56), (62, 66)\}$ obtained from them. Following, those values were denoted as d_{jk} and rewritten in the following form of points $(d_{jk}, 0)$. The points (landmarks and distances) were concatenated to form two vectors $v_{\text{upper}}, v_{\text{lower}}$, corresponding to the upper and lower face. A different way to represent the geometric variables is to 'flip' the data organization into curves but keep the control points organized with a distinctive pattern. Here each spatial coordinate $v_{\{\text{upper}, \text{lower}\}}$ will be used to smoothly interpolate and define a Hermite curve $u(v_{\{\text{upper}, \text{lower}\}})$ [34]. A Hermite curve uses the Hermite method of interpolation (derived from the calculation

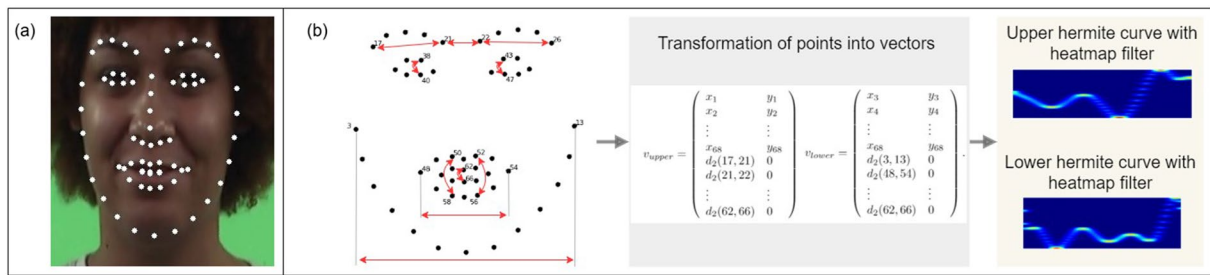


Fig. 5 A prototypical face mask is presented in (a) with white points $p_i = (x_i, y_i)$, $i = 1, \dots, 68$. In (b) is shown the geometrical features extraction, starting with the dots corresponding to the facial landmarks and the red lines connect the chosen landmarks locations for calculating geometric distances d_2 among them. Such measurements estimate the configuration of eyebrows, eyes, mouth, and cheeks. In

of divided differences algorithm) to obtain a continuous function, where numeric data are interpolated by piecewise polynomial specified at given values [64]. This method assumes uniform parameter spacing and is a particular case of a cardinal spline. Also, it is extensively used to obtain graphical curves, trajectories, and unveiled patterns in data points. Assembled from a Hermite curve sequence, the primary curve u is obtained by calculating tangents automatically from the user's control points. To generate the curve, a Hermite is drawn between each pair of points using the tangent calculated by:

$$R_i = \frac{(p_i - p_{i-1}) + (p_{i+1} - p_i)}{2}, \quad i = 1, \dots, 68$$

The curve passes through the user-defined points (minus the first and the last). For each segment's calculation, we use P_1 and P_4 , which are the start and end points; R_1 and R_4 , which are the tangent vectors at each point. This matrix will be recalculated and used in Hermite to interpolate each of the segments defined by a pair of points P_{i-1}, P_{i+1} . Thus, the formulation of a segment i of the curve is:

$$Q_i(t) = T_i M G$$

where the matrix $T_i = [t^3, t^2, t^1, t^0]$ is invariant, the base matrix $M = [P_1, P_4, R_1, R_4]$ with the point and tangent rule, and geometry matrix G is given by the control points, therefore, curve parameters. In other words, $Q_i(t)$ is a weighted sum of the elements of G by resolving matrix multiplication.

While this graphical curve can empower a lot of intelligence in the design and the model, it can also prove to be restrictive. These constraints can slow down the implementation process.

In addition, we applied a heatmap filter with a Gaussian coefficient of 16 in the trace of the curve u to turn the pattern more visually prominent (see Fig. 5). By generating such an intensity image graph, providing visual paths

the next box, these measures are used in the creation of the vectors v_{upper} and v_{lower} . These coordinates are interpolated to form the Hermite curve $u(v_{upper, lower})$. The last box shows an example of the final step of our geometric features where a heatmap filter with a gaussian coefficient of 16 is applied to the obtained Hermite curve

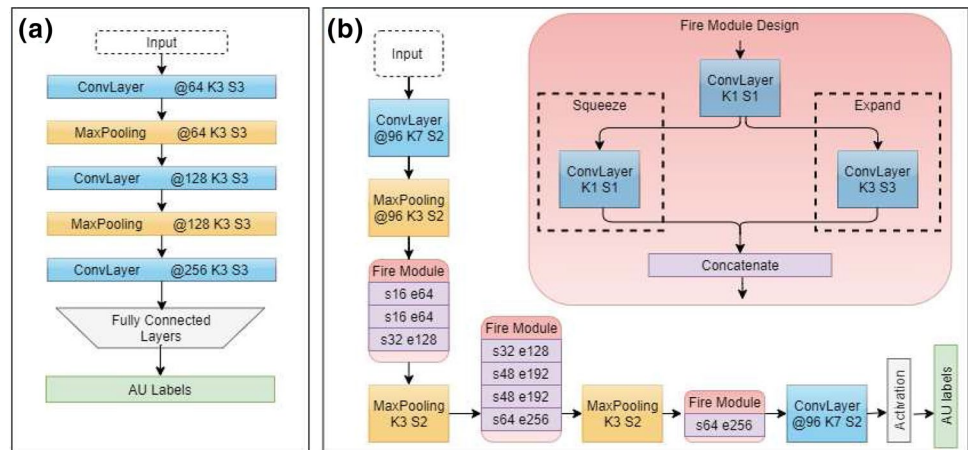
to understanding numeric values, it does faithfully preserve feature information of the geometrical points. Finally, the figure is resized to 36×96 and concatenated with the cropped image, forming two entry image of 96×96 size to the network model as presented in Fig. 4. Such image dimensions do not lose information. It is known that if the original images are too tiny, the performances of traditional FER methods would be unsatisfactory after image transformation. Our feature-based work exploited deep learning capabilities to compute the necessary information without the need for an explicit large image, particularly after the handcrafted creation of an input feature vector. This means that, ideally, the framework should focus only on essential parts of the face and should be less sensitive to other facial regions. To achieve this, most recent works that treat deep learning-based super-resolution algorithms would only include a single-cycle CNN such as [3,9]. Also, as the network depth increases, the number of parameters increases as well. Large-capacity networks will consume enormous storage resources and face overfitting problems. Therefore, adding multiple cycles from the horizontal process is preferable instead of deepening the network or even augmenting the entry image and encountering memory problems.

It is worth mentioning that the use of geometric information is applied in many SLR approaches. However, we use this information in a new way by transforming this data into curves and, later on, heatmap graphs. In other words, our contribution in the preprocessing phase is the way chosen for transforming the information taken from the points of the face.

4.3 AU detection model ensemble

We evaluated two backbone networks to implemented our method: model one is a basic CNN [61], and model two is the framework of SqueezeNet [25]. We chose to start with a CNN to be the first step toward a more robust network

Fig. 6 Input image, CNN architecture and SqueezeNet architecture. In image (a), we present the architecture of our CNN, consisting of one input layer, three convolution layers, two max pooling layers, and two full connection layers. In image (b), we present the structure of the SqueezeNet, where @,K,S,e,s denote number of filters, kernel size, stride, expand filters, and squeeze filters, respectively. Detailed descriptions are given in the text



model and is a widely applied model for performance comparison. In the case of SqueezeNet, to our knowledge, it has not been applied to AU recognition despite have shown their outstanding applicability and performance results in others computer vision application tasks (for instance, in pose recognition, object detection, among others). Such an ensemble of combined pre-processing of the region of interest and transformed geometric information with the SqueezeNet model for recognizing facial action units used in Libras discourse is a novelty contribution of this work. The architecture of each network is shown in Fig. 6.

The CNN model is designed from the concept that the architectures of a shallow CNN that can outperform modern deep CNNs [46, 49]. We use the information that a CNN with five hidden layers is already able to learn high-level discriminatory features to design our network. By having fewer layers and small convolution kernels size, the network has fewer parameters, which may take less time to train, without compromising its achievement of classification rates [37]. The model consists of a CNN where the first layer is a convolutional layer; the second layer is a max-pooling layer, the third layer is a convolutional layer, the fourth layer is another max-pooling layer, the fifth layer is a flatten layer, the sixth and the last layer are dense layers. The activation functions are all set to Rectified Linear Functions (ReLU).

We further ensemble the two networks (CNN and SqueezeNet) with the preprocessing stage mentioned above. Note that we will have two exits for each backbone network due to our preprocessing stage, relative to the face's upper and lower regions. Also, there are individual AUs and combinations of AUs; in fact, the head movement's AUs comes from the combination present in both regions. The method's output scheme is shown in Fig. 7. Then we test both

backbones in the same setup¹ to choose the one with the best results.

5 Datasets and evaluation methods

5.1 Data collection

We discuss the applied database containing FACS and are widely used in FER for deep learning algorithms evaluation. Our AU recognition task has not incorporated any other SLR data since no FACS annotated information is available in the literature. The proposed frameworks were trained using three datasets that had facial images with action unit labels being two of them composed with images from Libras' discourse:

HM-Libras dataset [11, 59] is a Libras' FACS annotated dataset, which includes 80 sequences from deaf individuals and sign language interpreters distributed under user-generated content on the internet. The set of videos has the presence of three women and seven men aging from 18 to 35, performing Libras' sentences of affirmative, WH-question, YN-question, and negative classes. AU annotations are provided in a frame-based manner. Each sequence starts and ends with a neutral expression. The metadata also include the Portuguese transcription. HM-Libras dataset is available for research purposes upon request to the authors.

SILFA dataset [58] is a dynamic video dataset of Libras' facial expressions of ten subjects (four women and six men), with ages from 18 to 30 and has Libras as their first language. A series of effective sentences are interpreted in Libras for facial action induction designed to obtain targeted facial expressions, including facial expressions that are not well documented in the sign language literature. The target

¹ Code available at <https://github.com/SrtaEmely/AUdetectionForLibras>.

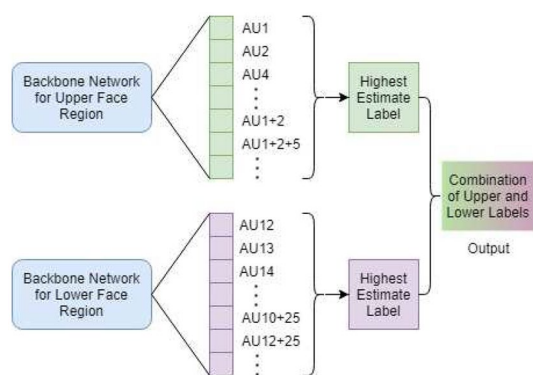


Fig. 7 Image illustrating the output of our proposed framework. The AUs' output label's concatenation is done by adding the outputs from two networks trained on the upper and lower regions

expressions include grammatical facial expressions (of a sentence, intensity, homonymy, and norm) and affective facial expressions (happiness, sadness, and anger). Each subject performed 23 sentences (total of 230 videos). Facial action units are annotated by frame-level, and the onset and offset of action units are coded. Also, there are annotations of grammatical facial expressions classes, as described in Table 1.

DISFA dataset [44] is a spontaneous dataset composed by videos of 27 subjects that vary in age from 18 to 50 years. The subjects are filmed while reacting to an emotional 4-min video stimuli. There are 12 AUs (1, 2, 4, 5, 6, 9, 12, 15, 17, 20, 25, and 26), labeled by two FACS coders. The lack of available data for comparing posed and spontaneous expression encouraged the same research group, to construct the Extended DISFA Dataset (DISFA +) [43], which contains the videos and AU annotations of posed and spontaneous facial expressions of 9 participants in the same format as DISFA.

5.2 Metrics

To evaluate the performance of the following experiments, we adopted the F1 frame-basic metric due to its capacity to describe the implementation of an algorithm in cases where samples are not balanced [40]. Widely used in AU detection, the F1 score is defined as the harmonic mean of precision and recall [9, 33, 78]. It was also calculated the average accuracy, precision, and recall (also called sensitivity) to explore more details of our model [59].

5.3 Comparative methods

For comparison, we selected three popular deep network architectures that were designed and trained on ImageNet, and have been successfully applied to multiple vision

problems: AlexNet [35], VGG-16 [63] and ResNet-50 [72]. To adapt both networks for our classification model, we modify the input and the output layer. The input layer was adjusted to accommodate images with a size of $60 \times 96 \times 3$. Also, the output layer was arranged to exit 32 labels for the upper part of the face and 85 labels for the lower part of the face. As a baseline, these networks were trained only with the ROI images, without geometric face information. Note that we do not use these pre-trained networks, as our evaluation is done for comparing architectures performance under the same conditions. Thus, the use of pre-trained networks is necessary to upscale the input images to the required sizes for these networks to work with pre-trained weights, making the experiment inappropriate.

5.4 Implementation details

Similar at what was done in [59], all architectures were based on Keras implementation [8]. We train every architecture for up to 300 epochs and a fixed mini-batch size of 32 samples. Both models were initialized with a learning rate of 0.01, optimizing the cross-entropy loss using stochastic gradient descent (SGD) with a momentum of 0.9 and weight decay of 0.001. Simard et al. [62] have shown that if the data are augmented in a reasonable way, the model can perform better. For training data augmentation, we use horizontal mirroring, randomly rotations and two types of shift and zoom transformations. These are applied randomly in each epoch creating twice the amount of data [59].

Due to the nature of the SL, we must take into account the occlusion of the face. When that happens we decide to drop the frame and mark with AU code 73. That happened in 3,4% of HM-Libras and 3% of SILFA datasets. In DISFA dataset was dropped 0.2% of samples mostly due to partial occlusions or extreme head rotations. Interestingly, both HM-Libras and SILFA are made from isolated sentences. Nevertheless, SILFA images have been elicited from Portuguese written sentences, while HM-Libras were obtained from natural SL discourse. Such complementary combination should add features to the analysis of the presence of facial expressions in SL discourse. They may, however, introduce a bias toward the grammatical structures of SL. Concerning the balance, the final contributions of samples can be visualized in Fig. 8.

Observe that our Libras samples in the datasets were combined with natural (conversation) and unnatural (scripted) sentences from discourse. This limitation may introduce bias, as the written form will influence the grammatical structure of SL. Such feature bias is present in many FER models. We are working toward diminishing their effects with more natural annotated samples, and it is understood that contributions could be made to expand the datasets for comparisons between users from other regions of Brazil.

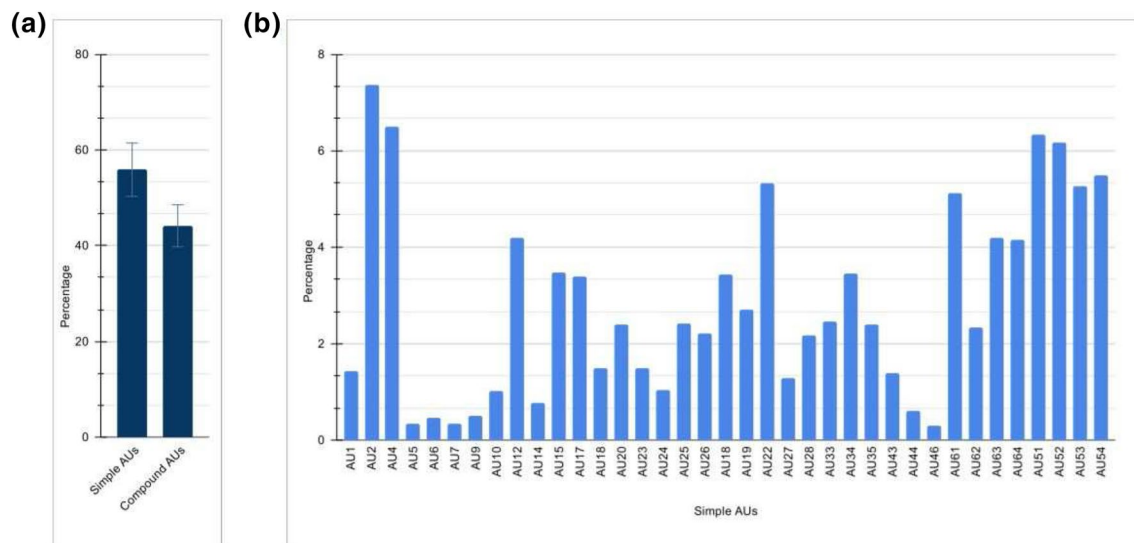


Fig. 8 Contributions in percentages of each of the AUs to the network. In image (a), we present the graph of the percentage composition of single and compound AU in our combined dataset (DISFA,

HM-Libras and SILFA). In image (b), we show the composition of simple AUs samples in the dataset

All experiments are performed on a PC with one NVIDIA GTX 1070 GPU. It took roughly 200 h to train each network until convergence that happened around 250 epochs.

6 Experiments and analysis

We simultaneously train two models to generate multi-class detection. Model one trained on the AU detection task of the upper face, and model two trained on the AU detection task of the lower face, which expects to learn sub-face level features and underlying AU correlation feature. We further ensemble the two networks with an average of their predicted probabilities according to 39 individual AU and 80 compounds FEs, then use the sigmoid layer normalizing the final result mapping in AU detection. The main advantage of using output compound AUs (which are combinations of individual AUs) is that there were AU combinations that do not appear alone, as in AU16 + 25 (very common in mouthings or spoken word). Of this technique, we hope that the single AU can be detected easily from the trained compounded one instead of merely allowing several individual AUs to be produced at once, enabling the creation of any compound AU from them.

In Sect. 6.1, the first experiment was done for our backbone network selection. We demonstrate the ensemble effectiveness in ablation experiments in Sect. 6.2. Comparison with other methods of AU detection in the literature is done in Sect. 6.3.

6.1 Backbone selection

We evaluated the proposed architectures, performing experiments with the datasets: DISFA, HM-Libras, and SILFA. We choose DISFA since it has AU annotations and is a standard dataset for AU detection, and to the best of our knowledge, HM-Libras and SILFA are the only Libras dataset with AU annotations. To ensure that the subjects are mutually exclusive in the train/test split sets, we adopt a form of the threefold partition by forming a group with DISFA, HM-Libras, and SILFA for training and using only a percentage of HM-Libras and SILFA for a test set. The training and testing protocol is performed by training on one set of data and testing on the other without any overlap. That way, we are able to demonstrate the effectiveness of the proposed model to the Libras application since extensive experiments have been conducted on our already described networks using a subject independent and cross-database approach. Table 2 demonstrates our experiment of backbone selection results following our described implementation details and evaluation strategy (Sects. 5.3, 5.4).

Analyzing Table 2 we can make a few observations. First, without the region constraint on AU and geometrical landmark detection, the baselines methods perform the worst. It is shown in Table 2 that our framework with SqueezeNet outperforms all other architectures, inclusive the benchmarks AlexNet, VGG-16 and ResNet-50. Compared with other works, the proposed framework with SqueezeNet achieves competitive performance with more AU labels, which demonstrates the effectiveness

Table 2 Comparative performance between backbone and benchmark networks

Architecture	Preprocessing	Metrics			
		Precision	Recall	Accuracy	F1
AlexNet	Upper face	0.7199	0.6226	0.6620	0.6306
	Lower face	0.8027	0.5719	0.5402	0.5454
	Avg.	0.76	0.59	0.60	0.58
VGG-16	Upper face	0.6123	0.6098	0.5322	0.5249
	Lower face	0.4732	0.4698	0.4082	0.3949
	Avg.	0.54	0.53	0.47	0.45
ResNet-50	Upper face	0.5768	0.4316	0.51	0.5066
	Lower face	0.4612	0.3535	0.4898	0.4542
	Avg.	0.52	0.39	0.50	0.48
CNN	Upper face	0.8564	0.6697	0.7999	0.7516
	Lower face	0.8187	0.6670	0.7257	0.7351
	Avg.	0.83	0.66	0.76	0.74
SqueezeNet	Upper face	<i>0.8956</i>	<i>0.8134</i>	<i>0.8991</i>	<i>0.8525</i>
	Lower face	<i>0.8840</i>	<i>0.8092</i>	<i>0.8635</i>	<i>0.8450</i>
	Avg.	0.85	0.71	0.88	0.84

Italic values represent the best results for each network

Bold values represent the best results overall

Table 3 Network performance evaluation according to the type of preprocessing scheme

Preprocessing type	Avg. Acc	Avg. F1
Raw image	0.4804	0.4952
Region of interest	0.6893	0.6629
Region of interest and geometry	0.7595	0.7728

of our proposed region and geometrical features learning. It can be observed that the this framework with SqueezeNet model significantly outperforms the first scheme with the CNN model. For that reason, we choose the SqueezeNet as the backbone for our Libras AU detection framework.

6.2 Ablation study

Raw data, as it is extracted from the camera, can be directly used as the characteristic vector to be used as input in deep learning algorithms. However, it is interesting to derive information with the potential to describe better the characteristics that mark each aspect of an FE. We hypothesized that by learning region of interest with geometrical features separately, a better understanding of AUs could be achieved.

Thereby, we propose an experiment with the preprocessing stage to focus on facial feature and action feature.

We separately pre-trained the SqueezeNet described in the Sect. 4.3 with three different feature extraction type of input: (1) the whole face image as extracted from the entry; (2) the face image separate into ROIs as describe in Sect. 4.1; and (3) the face image split into regions with geometric features as described in Sect. 4.2. All experiments were done in a threefold setting combining SILFA, HM-Libras, and DISFA datasets, where training was done up to 300 epochs. If convergence occurs before that, the training stops at the best result. Table 3, shows the average accuracy and F1 measure for AU recognition based in each type of preprocessing and step by step progress in result.

Table 3 shows a significant improvement in the classification rate when compared between raw image entry and ROI entry with geometric measures. These results can be attributed to the focus on the image information area, which cause to the network to benefit more. Also, larger images produce large feature maps, which has a negative impact on high-level features. Finally, the composition of ROI and geometric features ensemble on top of a SqueezeNet provides better average rates, accurate guidance for the network, and is the method chosen for our Libras' AU detection application.

6.3 Comparison with relevant state-of-the-art methods

We compared the performance of the proposed approach, referred to as Libras-SqueezeNet, against other relevant methods on DISFA dataset. These methods include Han et al. [23], AlexNet [35], Li et al. [38], EAC-Net [40], Mei et al. [45], Sankaran et al. [52], ARL [56], and STRAL [57] which report results on a subset of the 12 action units. To benchmark the performance of the Libras-SqueezeNet, we conduct experiments on this dataset under the same division strategy of the previous papers, which is the subject-exclusive threefold cross-validation setting. Note that most of these approaches use outside training data or additional facial landmark labels, while our method differs in training by using HM-Libras and SILFA dataset with more AU labels. The results are shown in Table 4. Such experiment was done so that we could compare our approach to other works in the field and gain further understanding on the learned features of the different AUs.

The presented values in Table 4 for the proposed framework Libras-SqueezeNet model achieve nearly close to state-of-the-art measures for some AU labels on DISFA. We can clearly see that our novel framework can adapt to various AUs with different variations, sizes, and non-rigid transformations for the AU detection task. Note that, for those AUs with very low occurrence rates, directly predicting them as non-occurrences can obtain poor individual accuracy but high average F1-frame results. Moreover, there is a more serious data imbalance issue in DISFA dataset, which

Table 4 F1 score result comparison with state-of-the-art methods on DISFA dataset

References	FACS-AUs										Avg.
	1	2	4	9	12	15	17	20	25	26	
AlexNet	0.12	0.12	0.28	0.12	0.30	–	–	–	0.44	0.28	0.24
EAC-NET	0.410	0.264	0.664	0.805	0.893	–	–	–	0.889	0.156	0.485
Han et al. [23]	0.437	0.400	0.672	0.497	0.758	0.378	0.523	–	0.724	0.548	0.55
Li et al. [38]	0.426	0.272	0.655	0.228	0.829	–	–	–	0.883	0.259	0.51
Mei et al. [45]	0.525	0.681	0.475	0.493	0.738	–	–	–	0.850	0.751	0.65
EvoNet	0.21	0.21	0.39	0.14	0.72	–	–	–	0.48	0.32	0.35
ARL	0.439	0.421	0.636	0.400	0.726	–	–	–	0.952	0.668	0.587
STRAL	0.522	0.474	0.689	0.567	0.725	–	–	–	0.913	0.676	0.63
Sankaran et al. [52]	0.448	0.417	0.529	0.507	0.724	0.377	0.458	–	0.822	0.609	0.491
Our CNN based	0.1	0.29	0.74	0.36	0.69	0.22	0.43	0.63	0.71	0.56	0.50
Libras-SqueezeNet	0.31	0.70	0.40	0.42	0.70	0.40	0.50	0.5	0.76	0.6	0.529

Table 5 Performance evaluation (F1-Scores) for 19 AUs on SILFA dataset

Reference	FACS-AUs									
	1	2	4	5	6	7	9	10	12	14
OpenFace	0.38	0.44	0.48	0.37	0.1	0.1	0.22	0.27	0.33	0.12
Ours CNN based	0.26	0.66	0.73	0.25	0.63	0.33	0.34	0.41	0.80	0.20
Libras-SqueezeNet	0.44	0.51	0.75	0.56	0.40	0.46	0.50	0.52	0.84	0.69
Reference	FACS-AUs									
	15	17	20	23	24	25	26	28	45	Avg.
OpenFace	0.34	0.15	0.18	0.22	–	0.43	0.41	0.19	0.40	0.285
Ours CNN based	0.80	0.34	0.57	0.69	0.26	0.84	0.82	0.72	0.42	0.530
Libras-SqueezeNet	0.88	0.65	0.60	0.72	0.28	0.87	0.88	0.74	0.41	0.615

results in large performance fluctuations over different AUs for most of the previous approaches [55], including our Libras-SqueezeNet.

One last comparison is made of our models' effectiveness by using an off-the-shelf AU regressor, the OpenFace [2]. Such trained analysis platform is capable of face detection and recognize the following subset AU 1, 2, 4, 5, 6, 7, 9, 10, 12, 14, 15, 17, 20, 23, 25, 26, 28, and 45. We set up an experiment on our test set (HM-Libras and SILFA), reporting results of 0.416 avg accuracy and 0.337 for F1-score for the eighty categories of AU. This model performed worse, which is likely because the multitask learning model has fewer parameters and learns fewer generalizations. In order to compare the performance in our dataset, when tested only for each AU component in OpenFace, we encounter a variation from 0.13 to 0.40, which are lower than the values obtained in our approaches. From these results, we conclude that ensembling state-of-the-art AU detection algorithms are most helpful for the excellent performance in a small set of AUs, but using region and geometrical features with combined AU is also helpful for a large number of AU labels.

6.4 Contribution of each component

Evaluations on SILFA dataset for the most common 19 AUs present in emotion studies [4, 16] are shown in Table 5. To have a fair comparison, we compare our frameworks to the reproduced performances for OpenFace [2]. For this experiment, we tested OpenFace and our approaches (our CNN based and Libras-SqueezeNet) on our SILFA dataset with the exact same splits. However, since OpenFace is not trained for a high variation of SL-related AUs, our reproduced performance scores are slightly lower than the published values for other datasets. From here, we can imply that training, including samples with head posing and a considerable variation of compound AUs, positively influenced our networks.

From Tables 4 and 5, we notice that the numerical results of our Libras-SqueezeNet on DISFA are lower than those on SILFA. This is due to the presence of unannotated AU in the DISFA image samples, while SILFA images are more challenging with all facial behavior annotated. Thus, our methods recognize AUs in the DISFA images that are not annotated, counting as a classification error.

Table 6 F1 score results for 20 AUs of different variants of our frameworks on SILFA dataset

Architecture	FACS-AUs									
	18	19	22	27	28	33	34	35	43	44
Ours CNN based	0.46	0.05	0.32	0.56	0.36	0.12	0.14	0.11	0.40	0.33
Libras-SqueezeNet	0.52	0.12	0.35	0.59	0.51	0.43	0.40	0.36	0.44	0.39
Architecture	FACS-AUs									
	46	61	62	63	64	51	52	53	54	Avg.
Ours CNN based	0.1	0.32	0.34	0.27	0.29	0.44	0.41	0.46	0.40	0.29
Libras-SqueezeNet	0.12	0.35	0.34	0.31	0.33	0.58	0.55	0.57	0.56	0.41

Furthermore, we use the proposed frameworks (our CNN based and Libras-SqueezeNet) to calculate the output scores contribution of each of the 14 AUs present only in this automatic facial expressions recognition for sign language study, as provided in Table 1. Table 6 brings the novelty in our approach by presenting the F1 score for the 14 AUs that in automatic facial expression and sign language recognition, to our knowledge, is completely ignored by researchers. Libras-SqueezeNet improves the estimates for each instance and the average F1-frame went from 29.4 to be 41.1 over our CNN based. Table 6 also shows that AU18, 27, and 28 have a better prediction overall. The lowest rate was AU19 (tongue show), which uses AU25, and the compound AU19 + 25 appears not to estimate as much the individual AUs as the individuals self-report measures of AU19, AU25, and AU26. Interestingly, while AU19 is considered to be among the more visually recognizable expressions and contributes to viseme classes, it did not achieve significant feature rates in this model on this dataset.

Note that AUs 61, 62, 63, 64 indicate eye gaze, while AUs 51, 52, 53, 54 indicate head pose, and they occur in most instances together within our datasets. For this reason, the estimation of one occurs at the expense of the other on our method. It is obvious that the system can detect head position with some degree of reliability, but the main advantage of this technique is that its possibilities of applications due to our vast number of AU detection for single components. Additionally, our method is non-intrusive and does not need user calibration as preprocessing.

Overall, the results presented in Table 6 show that our frameworks bring light to AU-related to SL, which is more vast than the number of AU recognized in other works that deal with AU emotion-related (presented in Sect. 6.3). The combined results here show that by exploring the benefit of the combined AU labels (multitask learning) in our proposed framework, the network can focus on both the AU regions and the related geometrical facial features, and thus is able to achieve better performance (an average F1-frame of 84%).

Our Libras-Squeezenet method was evaluated in both simple and compound AU presence from SILFA dataset and

is illustrated in Fig. 9(a). As one can see, the ICC is higher for compound AUs.

In our results, we also can see that AUs 2, 4, 12, 23, 25, 26, and 43 have higher accuracy than other AUs in Fig. 9b. This is because the prediction results of other AUs with occurrence rates far below 45% often strongly bias toward non-occurrence to cause low accuracy and F1-frame. Note that there are exceptions for AU 1, 5, 14, 23, 25, 43, and 46 have higher ICC rates. This demonstrates that correlations also influence the performance of different AUs among AUs. Most of the misclassification come from confusions between similar AUs: 1, 2, 23, 33, 34, 25, 26, 27, 43, and 45. The strong correlation between them caused confusion prediction between AU 1 and AU 2, mostly because the eyebrow raise brings both the brow's inner and outer portion into movement. The action of AU 33, which is the action of cheeks blowing, tends to pull the lips together as in AU 23 and AU34. Both AU 43 and AU 45 lower the eyelids and are often confused by human AU coders as well. All the mistakes of AU 26 were due to confusion with AU 25 and AU 27 since they all contain parted lips but differ only concerning the jaw's motion. Further comparison can be obtained by analyzing Figs. 8 and 9.

Due to lack of representation in training data and the application's inherent difficulty, the recognition of certain AUs is not as reliable as that of others. This area is still under active development and will continue to be refined with time, especially as more annotation in our datasets becomes available.

7 Conclusion

In this paper, we have studied an approach of using AU detection to predict Libras' facial expressions with outstanding AU generability recognition for new assignments. These would give invaluable information to analyze the language, which in turn could help with better engagement on automatic sign language recognition and a higher detection rate.

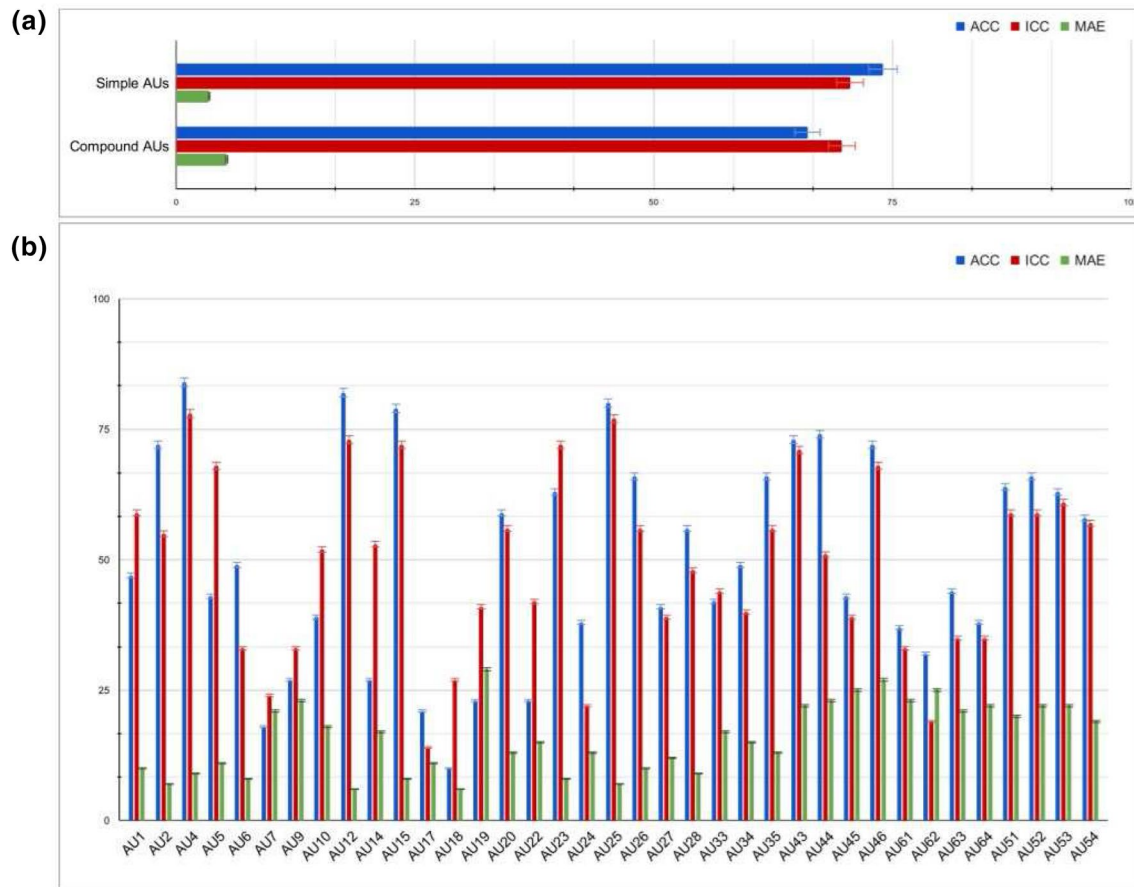


Fig. 9 Bar graphs showing the ACC (accuracy), MAE (Mean Absolute Error, where shorter bars mean better performance), and ICC (Intraclass Correlation Coefficient to measure the consistency

between members of each class), of the Libras-SqueezeNet method predicting AUs using frame-level labels on SILFA dataset

Summarizing, we try to innovate the premise of the necessity of having a unified feature representation. This was done by concatenating region of interest and geometric features. We theoretically and experimentally show that learning such a preprocessing stage in combination with SqueezeNet can be an alternative to existing methods. Also, we apply our framework to multiple modalities of Libras datasets: HM-Libras and SILFA. Additionally, our AU detection method has the capability to predict 39 simple FEs and around 80 combinations of AUs FACS, which is more than the average number of AU FACS encountered in the facial action detection literature.

However, there are limitations to our ability to successfully predict Libras' facial expressions in certain conditions. For example, having significant head movement or hand obstructions on their faces limited the performance of the models. The geometric features refined the

performance, although a better way to improve accuracy in such conditions is to train on a broader range of images that include various head poses and facial obstructions by hand movements. Future work could entail training with additional datasets and using more advanced deep learning techniques, such as combining models that would know prior predictions, which could improve performance.

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